

as augmented reality (AR), chip design, and advanced manufacturing. In China, there is a hospital that operates entirely with robots, eliminating the need for human intervention. In the US, banks like Bank of America are reducing the number of physical branches due to AI-driven innovations.

Real-time systems and edge computing are also transforming development. Instead of sending all data to the cloud, processing at the edge saves time and resources. Business logic and data remain local, with only critical, aggregated insights transmitted to the cloud.

The demand for quick, cost-effective innovation has made edge computing and smart IoT essential in Industry 4.0. As roles evolve, continuous learning remains key. It is essential to remain curious and adapt to the evolving technological landscape.

The journey of 75 years...

The timeline of AI begins in 1950. Although the pioneers of that era could not have imagined the advances seen today, their work laid the foundation for what exists in 2025—including tools like ChatGPT. Where it goes in the next 75 years remains to be seen. But one thing is certain: evolution is constant, and change is inevitable.

Many of us may have seen the familiar AI timeline diagram across platforms. At its core, AI is software designed to mimic the human brain. Machine learning is a subset of AI, with deep learning as a further subset. Within deep learning lies generative AI, where large language models (LLMs) represent the most significant current development.

Over the last decade, the evolution of generative adversarial networks (GANs) has been remarkable. GANs have played a crucial role in enabling the creation of LLMs. The technology has progressed rapidly since 2013, driven by breakthroughs in deep learning.

Three key concepts must be remembered as the field progresses: generative adversarial networks, large language models, and transformers. A well-known research paper by Google, titled *Attention is All You Need*, laid the foundation for transformer models—technologies that continue to shape the future of AI.

What is generative AI

This is where the development of transformers began, pioneered by engineers at Google. Consider this: Netflix took 3.5 years to reach one million users. ChatGPT achieved the same in just five days. Several other companies also contributed to this

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user milestone over the years. We must understand why ChatGPT was able to scale so quickly—this lies in the power of generative AI.

There are two major types of AI: discriminative and generative. Discriminative AI is used for classification tasks, such as identifying spam emails, through supervised learning. It enables accurate decision-making via structured learning paths.

Generative AI, however, goes further. It not only classifies but also creates text, images, and videos based on vast amounts of data. These models are trained on massive datasets sourced from public domains, including Facebook, LinkedIn, Instagram, and Twitter.

Generative models like GPT are trained using approximately 75 billion parameters, not just data

points but fine-tuned variables that shape their output. Hence, the name: Generative Pre-trained Transformers.

These models are already trained and ready to use. OpenAI launched its first product in January 2023, and version 4.0 is now available as a commercial solution widely considered for adoption across various industries.

Roadmap for beginners

Many fresh engineers are eager to enter the AI field, and even experienced professionals are considering the transition. Not long ago, a programme manager with 25 years of experience expressed interest in switching to AI. While this is a wise career move, it requires careful planning, especially at a stage of life when responsibilities tend to be greater. One cannot simply quit everything overnight; it is important to balance the current role with steady progress. Start small and be consistent.

Ask three key questions: What, How, and Why? The 'Why' is crucial—why invest time in learning, or attend a conference on a weekend? Because it adds value to one's career. That self-reflection is the first step, and each step builds upon the previous one.

For beginners, establishing a solid foundation in basic programming and microcontrollers is essential. Relying solely on tools like ChatGPT to write and run code without understanding the underlying logic is not advisable. Learn the fundamentals, such as the architectural differences between microcontrollers like the 8086 and 8051, and then attempt small projects.

I can share a personal anecdote here. My 10-year-old son built a simple IoT project called "Water is Life" using Arduino. It measured humidity across different plants in our home and alerted us when watering was needed. He compiled the code independently and submitted it to a competition,